CHAPTER

SIX

Atomic Spectra

- Questions on Chapter Six from the Previous Years Exams.
- Test on Chapters 5 and 6

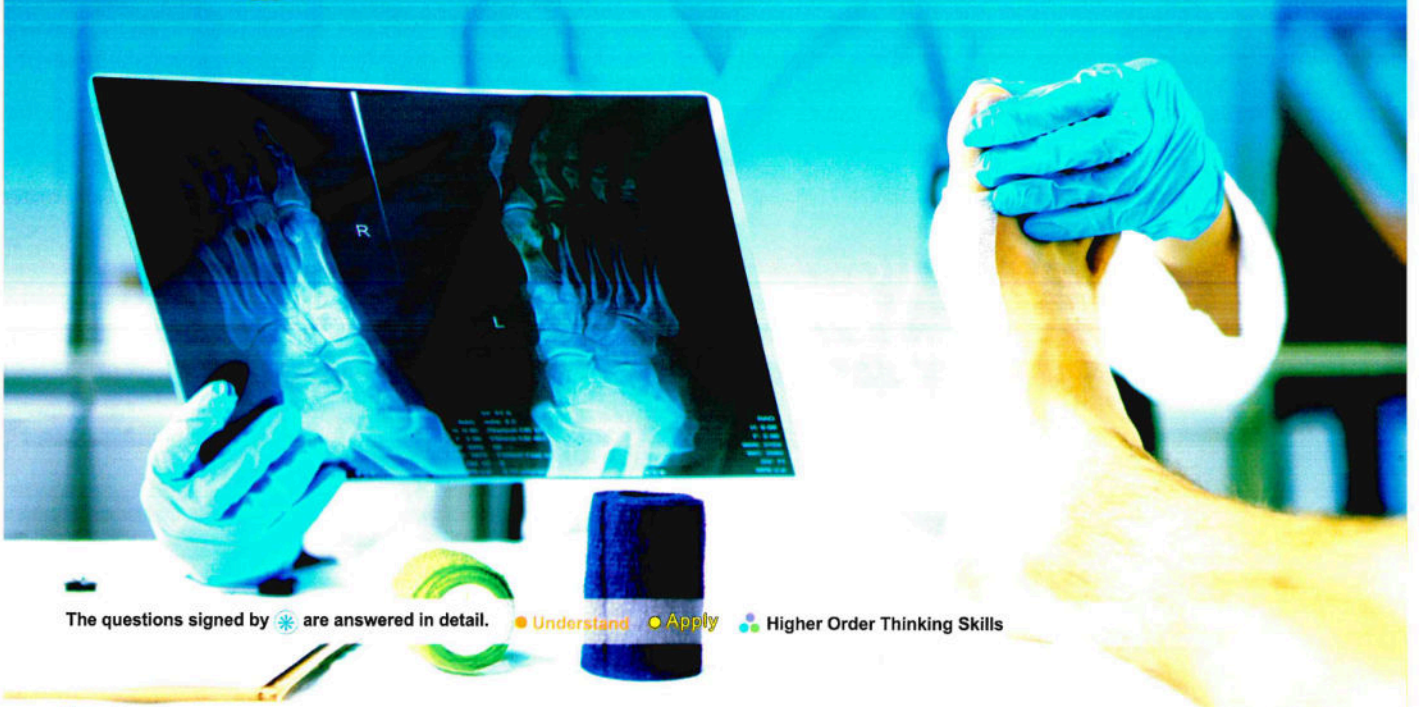


Chapter 6

Questions on Chapter Six

Atomic Spectra

Interactive test



The questions signed by ✳ are answered in detail. ● Understand ○ Apply ⁂ Higher Order Thinking Skills

Important guidelines

Hydrogen atom

Radius of the orbit

$$r = \frac{n\lambda}{2\pi} = \frac{nh}{2\pi m_e v}$$

The difference in energy between two levels

$$\Delta E = E_{higher} - E_{lower}$$

Energy of the level

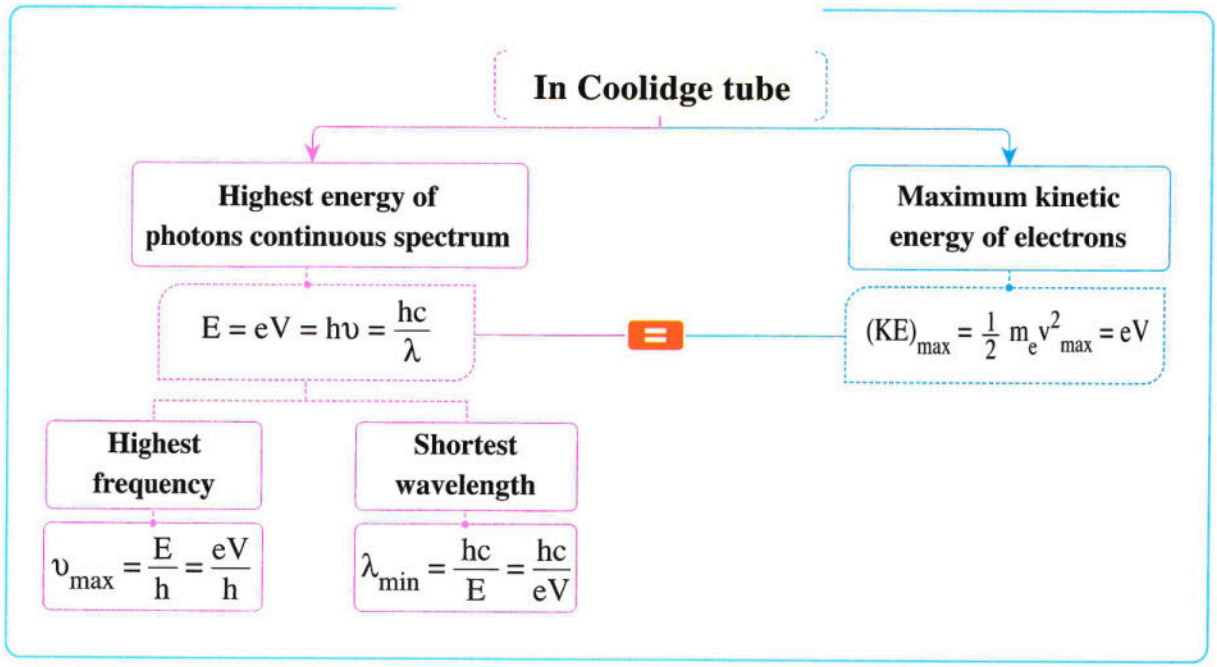
$$E_n = -\frac{13.6}{n^2}\ (eV)$$

Highest energy in the series (highest frequency - shortest wavelength)

$$E_\infty - E_n = \frac{hc}{\lambda_{min}} = h\upsilon_{max}$$

Lowest energy in the series (lowest frequency - longest wavelength)

$$E_{n+1} - E_n = \frac{hc}{\lambda_{max}} = h\upsilon_{min}$$



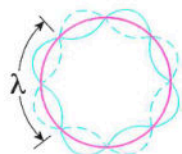
First Multiple choice questions

Use the following constants if needed:

$e = 1.6 \times 10^{-19}$ C, $m_e = 9.1 \times 10^{-31}$ kg, $h = 6.625 \times 10^{-34}$ J.s, $c = 3 \times 10^8$ m/s

Bohr's model

1. An electron moves in energy level (n = 4) around the nucleus of hydrogen atom associated with a standing wave of wavelength λ as in the opposite figure, so the radius of the energy level (r) is ..... .



(a) $\frac{4\lambda}{\pi}$ (b) $\frac{2\lambda}{\pi}$ (c) $\frac{\lambda}{\pi}$ (d) $\frac{\lambda}{2\pi}$

2. According to Bohr's model, if the wavelength of the wave that is associated with the electron motion in one of the energy levels of the hydrogen atom is equivalent to πr where (r) is the radius of the orbit in which that electron is located, then that electron is rotating in the energy level ..... .

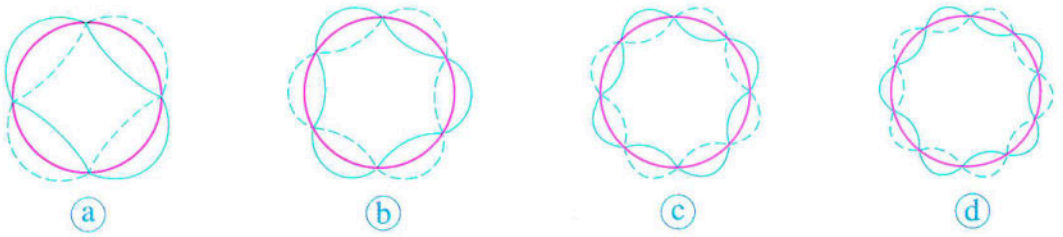
(a) K (b) L (c) M (d) N

3. Which of the following values could represent the wavelength associated with the motion of an electron in a hydrogen atom, given that (r) is the radius of the atom ?

(a) $5\pi r$ (b) $3\pi r$ (c) $2\pi r$ (d) $\frac{4}{3}\pi r$

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4 If the wavelength associated with the motion of an electron in a certain orbit in the hydrogen atom is 13.32 Å and the circumference of this orbit is 53.3 Å, so according to Bohr's model, which of the following diagrams shows the standing wave which is associated with the motion of the electron in this orbit?

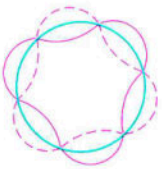


(a) (b) (c) (d)

5 ✱ An electron moves with a speed of 7.28×10^5 m/s in the third orbit of the hydrogen atom, then the radius of the third orbit in the hydrogen atom equals ..... .

(a) 4.77×10^{-10} m (b) 9.54×10^{-10} m (c) 1×10^{-9} m (d) 47.7×10^{-9} m

6 ✱ The opposite figure represents a standing wave associated with an electron in one of the energy levels of the hydrogen atom according to Bohr's model, then:



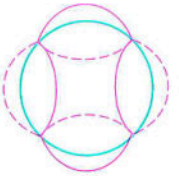
(i) The orbit order (n) in which this electron is found from the nucleus is equal to ..... .

(a) 1 (b) 2 (c) 3 (d) 4

(ii) If the radius of the level in which the electron is found is equal to 4.761×10^{-10} m, the wavelength of the standing wave associated with the electron equals ..... .

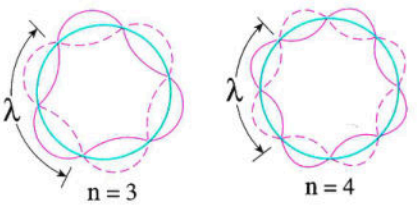
(a) 1.5×10^{-9} m (b) 3×10^{-9} m (c) 7.49×10^{-10} m (d) 9.98×10^{-10} m

7 ✱ The opposite figure shows a standing wave associated with the electron motion in one of the orbits of the hydrogen atom, so when the speed of the electron in that orbit is 1.09×10^6 m/s, then the radius of the orbit equals ..... .



(a) 1.06×10^{-10} m (b) 2.13×10^{-10} m (c) 4.25×10^{-10} m (d) 6.68×10^{-10} m

8 ✱ The opposite figure represents two energy levels in the hydrogen atom, then the ratio between the speeds of the electron in the two cases in terms of the radius of the orbits $\left(\frac{v_3}{v_4}\right)$ is ..... .

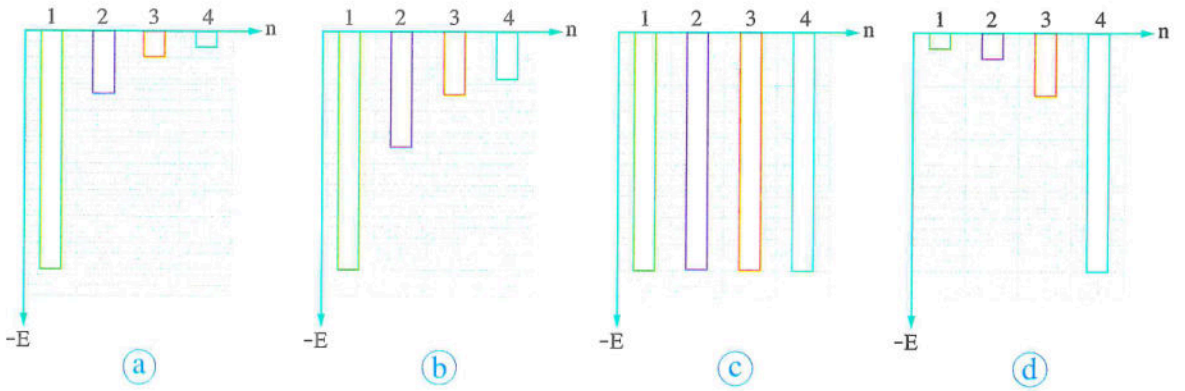


(a) $\frac{3\,r_3}{4\,r_4}$ (b) $\frac{4\,r_3}{3\,r_4}$

(c) $\frac{3\,r_4}{4\,r_3}$ (d) $\frac{4\,r_4}{3\,r_3}$

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9 Which of the following graphs represents the relation between the energy of the level (– E) and the level order (n) of the hydrogen atom according to Bohr's model?



10 An electron becomes stable in the ground energy level when it ..... .

(a) acquires energy continuously (b) acquires quantized energy

(c) doesn't gain energy (d) loses its energy continuously

11 An electron moves from a lower energy level (E_1) to a higher energy level (E_2), when the atom ..... .

(a) absorbs a photon of energy equals ($E_2 - E_1$)

(b) emits a photon of energy equals ($E_1 - E_2$)

(c) absorbs a photon of energy equals ($E_1 + E_2$)

(d) emits a photon of energy equals ($E_1 + E_2$)

12 When an electron in the hydrogen atom moves from a level of energy –3.4 eV to another of energy –13.6 eV, this means that the hydrogen atom has ..... .

(a) absorbed a photon of energy 10.2 eV (b) absorbed a photon of energy 17 eV

(c) emitted a photon of energy 10.2 eV (d) emitted a photon of energy 17 eV

13 The electron of hydrogen atom orbits in the first energy level (K), so:

(i) The least energy that releases an electron from the atom equals ..... .

(a) 13.6 eV (b) 10.2 eV (c) 0.85 eV (d) 3.4 eV

(ii) The largest wavelength for a photon that is absorbed by a hydrogen atom and leads to its ionization equals ..... .

(a) 9.1×10^{-8} m (b) 8.4×10^{-8} m (c) 8.1×10^{-8} m (d) 8.6×10^{-8} m

14 The minimum quanta of energy required to excite a hydrogen atom from the ground state equals .....

(a) 13.6 eV (b) 6.8 eV (c) 10.2 eV (d) 3.4 eV

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15 An electron in a certain atom has moved from a higher energy level E_2 to the lower energy level E_1, so the wavelength of the emitted photon can be determined from the relation; ..... .

(a) $\frac{E_2 - E_1}{hc}$ (b) $\frac{hc}{E_2} - \frac{hc}{E_1}$ (c) $\frac{hc}{E_2 - E_1}$ (d) $\frac{c}{h(E_2 - E_1)}$

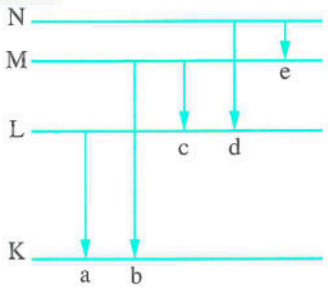
16 The longest wavelength in Lyman series results from the transition of an electron between levels ..... .

(a) $n = 3 \longrightarrow n = 2$ (b) $n = \infty \longrightarrow n = 2$ (c) $n = \infty \longrightarrow n = 1$ (d) $n = 2 \longrightarrow n = 1$

17 The maximum frequency of a photon emitted in Brackett's series results from transition of the electrons between levels ..... .

(a) $n = 5 \longrightarrow n = 4$ (b) $n = 4 \longrightarrow n = 3$ (c) $n = \infty \longrightarrow n = 4$ (d) $n = \infty \longrightarrow n = 3$

18 The opposite figure represents possible transitions for an electron in a hydrogen atom, so the photon resulted from:



(i) Transition (a) its energy equals ..... .

(a) 2.2×10^{-18} J (b) 1.9×10^{-18} J
(c) 1.6×10^{-18} J (d) 1.1×10^{-18} J

(ii) Transition (b) its frequency equals ..... .

(a) 1.5×10^{15} Hz (b) 2.1×10^{15} Hz (c) 2.5×10^{15} Hz (d) 2.9×10^{15} Hz

(iii) Transition (c) its wavelength approximately equals ..... .

(a) 328 nm (b) 657 nm (c) 948 nm (d) 984 nm

(iv) Transition (d) its momentum equals ..... .

(a) 5.44×10^{-27} kg.m/s (b) 2.72×10^{-27} kg.m/s
(c) 1.36×10^{-27} kg.m/s (d) 1.12×10^{-27} kg.m/s

(v) Transition (e) its equivalent mass equals ..... .

(a) 1.2×10^{-36} kg (b) 2.4×10^{-36} kg (c) 3.6×10^{-36} kg (d) 4.8×10^{-36} kg

19 The shortest wavelength in Balmer series approximately equals ..... .

(a) 3943 Å (b) 3850 Å (c) 3653 Å (d) 3450 Å

20 The shortest wavelength in Pfund series equals ..... .

(a) 21652 Å (b) 22834 Å (c) 23161 Å (d) 23558 Å

21 * In the spectrum of the hydrogen atom, the ratio between the longest wavelength in Lyman series to the longest wavelength in Balmer series is ..... .

(a) $\frac{5}{27}$ (b) $\frac{3}{23}$ (c) $\frac{7}{27}$ (d) $\frac{9}{31}$

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22 * The ratio between the longest wavelength to the shortest wavelength in Lyman series for the spectrum of the hydrogen atom equals ..... .

(a) $\frac{25}{9}$ (b) $\frac{17}{6}$ (c) $\frac{9}{5}$ (d) $\frac{4}{3}$

23 If the shortest wavelength in Lyman series is λ, then the shortest wavelength in Balmer series is ..... .

(a) $\frac{\lambda}{4}$ (b) $\frac{\lambda}{2}$ (c) 2λ (d) 4λ

24 * In the hydrogen atom, if υ_1 is the minimum frequency in Paschen series and υ_2 is the minimum frequency in Balmer series, then the ratio $\left(\frac{\upsilon_1}{\upsilon_2}\right)$ is ..... .

(a) $\frac{7}{20}$ (b) $\frac{4}{3}$ (c) $\frac{20}{7}$ (d) $\frac{5}{2}$

25 * In the hydrogen atom, if the energy of the second level is (–E), the energy of the third level is ..... .

(a) $-9\ E$ (b) $-\frac{E}{9}$ (c) $-\frac{4}{9}E$ (d) $-\frac{9}{4}E$

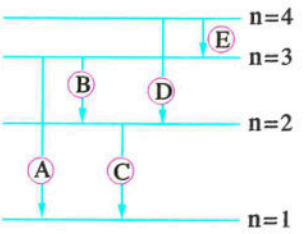
26 If the De-Broglie wavelength which associates with the motion of an electron in a certain energy level of radius r_n in a hydrogen atom equals $\frac{2\pi r_n}{5}$, the minimum energy that should be acquired by the electron to completely leave the atom equals ..... .

(a) 0.544 eV (b) 0.942 eV (c) 2.72 eV (d) 3.4 eV

27 In the hydrogen atom, the wavelength of the emitted radiation due to the transition of an electron from energy level of order n to the first energy level is 9.51×10^{-8} m. If the energy of the first level is -2.176×10^{-18} J, n equals ..... .

(a) 6 (b) 5 (c) 4 (d) 3

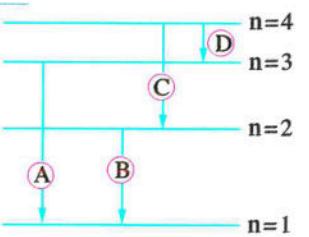
28 The opposite figure represents several transitions (A), (B), (C), (D) and (E) of an electron in the hydrogen atom between the energy levels, which one of these transitions gives a spectrum line that lies in Balmer series?



(a) (A) and (B). (b) (A) and (C).

(c) (E) only. (d) (B) and (D).

29 The opposite figure represents four transitions of an electron in the hydrogen atom between the energy levels, which one of the following statements is correct?



(a) The transition (D) gives a spectrum line has the least wavelength.

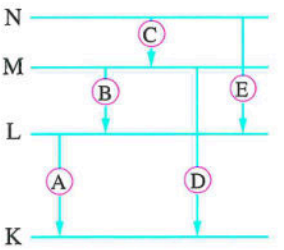
(b) The transition (C) gives a spectrum line in the ultraviolet rays region.

(c) The transition (B) gives a spectrum line in the infrared rays region.

(d) The transition (A) gives the highest frequency between these transitions.

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30 The opposite diagram represents multiple transitions (A), (B), (C), (D) and (E) of an electron in the energy levels of a hydrogen atom, so which of the following statements is **not** correct?



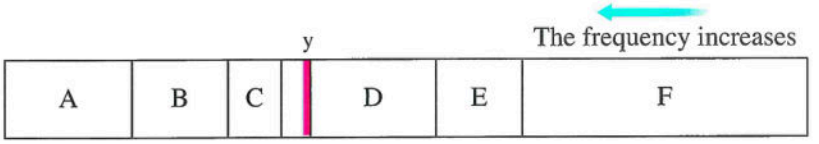
(a) Transition (C) produces a line spectrum in the IR region.

(b) Transition (D) produces the shortest wavelength of these transitions.

(c) Transition (E) produces the highest frequency of these transitions.

(d) Transition (B) produces a line spectrum in the visible light region.

31 The following figure represents the regions of the electromagnetic spectrum, if line y represents a red spectrum line emitted from the hydrogen atom :



In which of the shown spectrum regions does the linear spectrum of hydrogen lie when an electron moves from the energy level (O) to the energy level (M)?

(a) Region A. (b) Region B. (c) Region C. (d) Region D.

32 The photon that can be absorbed by a hydrogen atom to be excited from the ground state to a highest level, belongs to ..... .

(a) radio wave region (b) the visible light region

(c) the ultraviolet region (d) the infrared region

33 When two photons (a) and (b), with energies 4.91 eV and 10.2 eV respectively, pass by a hydrogen atom in its ground state, which of the two photons can be absorbed by the atom and participate in excitation ?

(a) Photon (a). (b) Photon (b). (c) Both photons. (d) Neither photon.

34 The ratio between the momentum of the photon emitted from Lyman series to the momentum of the photon emitted from Balmer series is ..... .

(a) equal to 1 (b) greater than 1 (c) less than 1 (d) indeterminable

35 If the possible number of energy levels for the electron movement in an atom is four energy levels and the electron can move between any two levels of these levels, the number of spectrum lines which can be emitted is ..... .

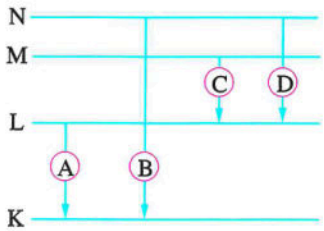
(a) 3 (b) 6 (c) 8 (d) 10

36 If the energy of the electron in one of the energy levels of the hydrogen atom is equal to –3.4 eV and the radius of this level is 2.13 Å, the De-Broglie wavelength which is associated with the motion of the electron in this level is ..... .

(a) 13.38 Å (b) 9.99 Å (c) 6.69 Å (d) 3.33 Å

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37 The opposite figure illustrates several possibilities for electron transitions in a hydrogen atom, where each of them leads to the emission of one of the wavelengths of the line spectrum of the hydrogen atom. So, which of the following choices is correct?



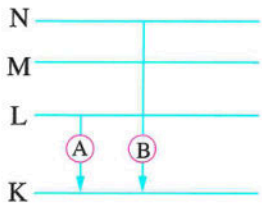
(a) $\lambda_A < \lambda_B$ (b) $\lambda_C < \lambda_D$

(c) $\lambda_D < \lambda_B$ (d) $\lambda_A < \lambda_D$

38 * If you know that the shortest wavelength in one of the hydrogen atoms spectrum series is 14610 Å , the name of this series and the longest wavelength of this spectrum are ..... .

	Name of the series	The longest wavelength
(a)	Paschen	9671 Å
(b)	Bracket	9671 Å
(c)	Pfund	60443 Å
(d)	Bracket	40594 Å

39 In the opposite figure, two possible transitions of the electron in the hydrogen atom, so the ratio between the wavelengths of the emitted photons during atomic transitions A and B ($\frac{\lambda_A}{\lambda_B}$) equals ..... .



(a) $\frac{E_L - E_K}{E_N - E_K}$ (b) $\frac{E_N - E_K}{E_L - E_K}$

(c) $\frac{E_L}{E_N}$ (d) $\frac{E_N}{E_L}$

40 * The opposite table shows energy of some energy levels in hydrogen atom. If an excited electron in energy level of order n has fallen to the first energy level and emitted a photon of equivalent mass 2.267×10^{-35} kg due to its transition from the energy level n, the value of n equals ..... .

Level	Energy of the level (eV)
K	– 13.6
L	– 3.4
M	– 1.51
N	– 0.85
O	– 0.544

(a) 2 (b) 3 (c) 4 (d) 5

41 When hydrogen atoms in the ground state are excited by a monochromatic electromagnetic radiation of photon energy 12.75 eV, the number of possible spectral lines that can be emitted :

(i) In the Lyman series, ...

(a) 1 (b) 2 (c) 3 (d) 4

(ii) In the Balmer series ..... .

(a) 1 (b) 2 (c) 3 (d) 4

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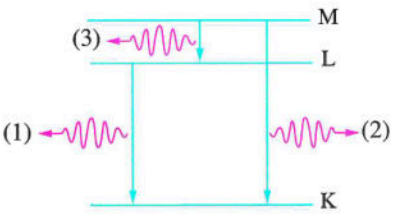
42 A hydrogen atom in the ground state is excited by a photon of a ray of wavelength 975 Å, so the excitation level and the number of the possible spectral lines that could be emitted when the atom relaxes, respectively, are ..... .

(a) L, 6 (b) L, 1 (c) N, 6 (d) N, 1

43 * When the electron moves from the fifth level to the second level, the wavelength of the produced radiation in hydrogen atom equals ..... . (Knowing that: $E_1 = -13.6$ eV)

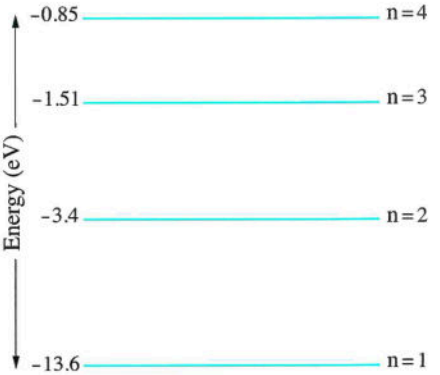
(a) 2283 Å (b) 4349.4 Å (c) 6959 Å (d) 12421.9 Å

44 * The opposite figure illustrates some electron transitions between energy levels in the atom of an element, if the wavelengths of the two photons (1) and (2) are 304.45 nm and 256.86 nm respectively, the wavelength of photon (3) equals ..... .



(a) 1643 nm (b) 1800 nm (c) 1928 nm (d) 2428 nm

45 * From the opposite figure, when an electron of hydrogen atom is in the fourth energy level, then:



(i) The number of possible emissions of photons of different frequencies in this case equals ..... .

(a) 4 (b) 6

(c) 8 (d) 10

(ii) The minimum frequency of a photon that can be emitted from the atom in this case is ..... .

(a) 1.3×10^{13} Hz (b) 9.96×10^{13} Hz (c) 3.08×10^{15} Hz (d) 1.59×10^{14} Hz

(iii) The maximum frequency of a photon that can be emitted from the atom in this case is ..... .

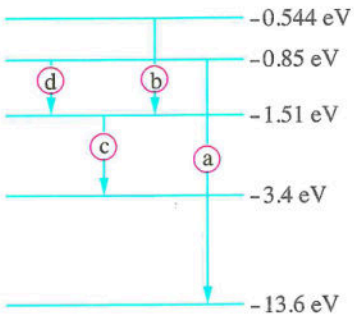
(a) 2.1×10^{13} Hz (b) 1.92×10^{14} Hz (c) 3.08×10^{15} Hz (d) 6.59×10^{15} Hz

46 * When the electron of the hydrogen atom is transferred from the third orbit to the first orbit, a photon is produced and collided with the cathode of a photocell, so an electron is emitted from the photocell cathode with a maximum kinetic energy of 9.79 eV, then the work function of the surface of the photocell cathode equals ..... .

(a) 2.1 eV (b) 2.3 eV (c) 4.3 eV (d) 4.7 eV

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47 * The opposite figure shows four transitions (a), (b), (c) and (d) for the electron of the hydrogen atom, which of these transitions produces a photon of wavelength 656 nm?



(a) Transition (a).
(b) Transition (b).
(c) Transition (c).
(d) Transition (d).

48 A photon of wavelength 486.3 nm is emitted from the hydrogen atom, then:

(i) The photon energy equals ..... .

(a) 2 eV (b) 2.55 eV (c) 3 eV (d) 3.55 eV

(ii) The electron transition is between ..... .

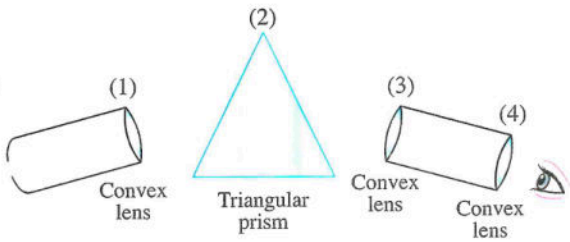
(a) the fourth and the first energy levels
(b) the fourth and the second energy levels
(c) the third and the first energy levels
(d) the third and the second energy levels

49 * An excited electron in the hydrogen atom moves from one of the higher energy levels (x) to the ground state in two successive stages, emitting photons of wavelength 2624 nm and 97.45 nm respectively, then the order of the energy level (x) of the excited electron is ..... .

(a) 2 (b) 3 (c) 5 (d) 6

The spectrometer and the spectra

50 The opposite diagram illustrates the optical components in the structure of a spectrometer, so the component that works on:



(i) Dispersing the spectra according to their wavelengths is ..... .

(a) (1) (b) (2) (c) (3) (d) (4)

(ii) Converging each light color into a certain focus on the focal plane is ..... .

(a) (1) (b) (2) (c) (3) (d) (4)

51 The emission spectrum of elements is produced due to transition of an electron in the atom from ..... .

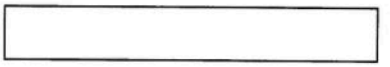
(a) a higher energy level to a lower energy level
(b) a lower energy level to a higher energy level
(c) the ground state level to the nucleus
(d) the ground state level to outside of the atom

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52 When the light of a tungsten bulb is analysed after passing through sodium vapor, we get ..... .

(a) colored lines on a dark background
(b) colored lines on a white background
(c) dark lines on a colored background
(d) continuous colored region

53 Which of the following figures represents the absorption spectrum of an element?

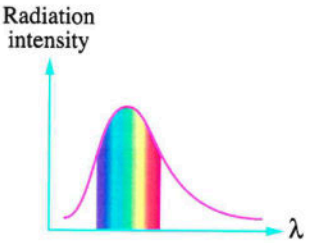
(a) 
(b) 
(c) 
(d) 

54 The opposite figure shows the spectrum which is obtained from a spectrometer, which of the following choices represents the source of that spectrum?



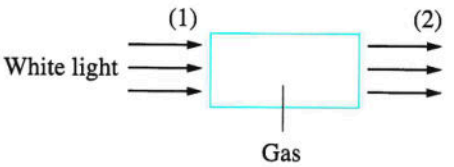
(a) Tungsten bulb.
(b) Neon lamp.
(c) Hot gas.
(d) White light after passing through a gas.

55 The opposite graph represents a spectrum of ..... .



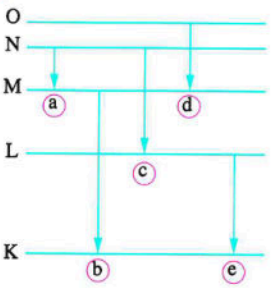
(a) continuous emission
(b) linear emission
(c) linear absorption
(d) monochromatic light

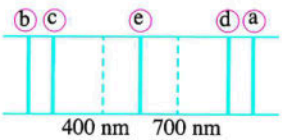
56 When white light (1) passes through a gas as in the opposite figure, then the produced spectrum (2) passes through a spectrometer, the resulted spectrum will be ..... .

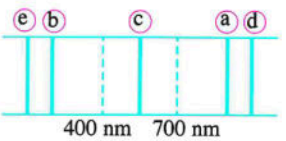


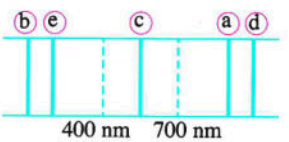
(a) a continuous spectrum
(b) a line emission spectrum
(c) dark lines on a bright background
(d) bright lines on a dark background

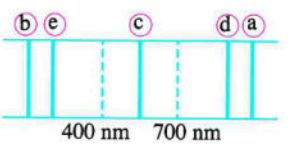
57 The opposite figure represents possible transitions (a), (b), (c), (d) and (e) of electrons in hydrogen atoms, which of the following choices represents the spectral line produced in spectrometer as a result of these transitions?



(a) 

(b) 

(c) 

(d) 

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X-rays

58 When X-rays pass perpendicular to a strong uniform magnetic field, ...

(a) they don't deflect from their path

(b) they deflect in a direction opposite to that of the field

(c) they deflect in a direction perpendicular to that of the field

(d) their speed increases

59 The ability of X-rays produced from Coolidge tube to penetrate bodies **does not** depend on ..... .

(a) the wavelength of the produced radiation

(b) the energy of the electrons which collide with the target

(c) the current intensity of the filament

(d) the potential difference between cathode and anode

60 The production of X-rays from Coolidge tube represents a model for the conservation of energy, so what is the correct order for the energy conversion starting from getting emitted from the filament till colliding with the target?

(a) Mechanical energy ⟶ Electric energy ⟶ Electromagnetic energy

(b) Electromagnetic energy ⟶ Mechanical energy ⟶ Electric energy

(c) Electric energy ⟶ Mechanical energy ⟶ Electromagnetic energy

(d) Electric energy ⟶ Electromagnetic energy ⟶ Mechanical energy

61 X-ray spectrum cannot be produced from an excited hydrogen atom because ..... .

(a) the energy of the level K is less than the energy of the photons of X-rays

(b) the energy of the level K is greater than the energy of the photons of X-rays

(c) the energy of the level L is higher than the energy of the X-ray photons

(d) the energy of the level M is higher than the energy of the X-ray photons

62 In Coolidge tube, if the potential difference between the terminals of the filament is doubled, the wavelength of the linear spectrum of X-rays ..... .

(a) increases to double

(b) decreases to half

(c) does not change

(d) increases 3 times

63 The wavelength of the characteristic X-ray spectrum depends on ..... .

(a) the electric current intensity which passes in the filament

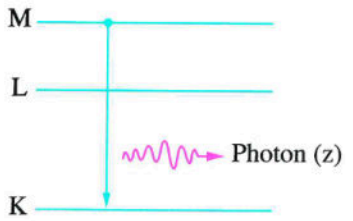
(b) the potential difference between the filament and the target

(c) the type of the target material

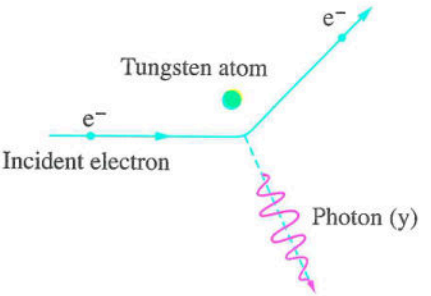
(d) the air pressure inside the tube

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64 Two X-ray photons (z and y) are emitted from a Coolidge tube, photon z is emitted due to the transition of an electron between two energy levels in a tungsten atom as shown in figure (1), while photon y is emitted due to the passage of an electron near the nucleus of the tungsten atom as in figure (2):



(1)

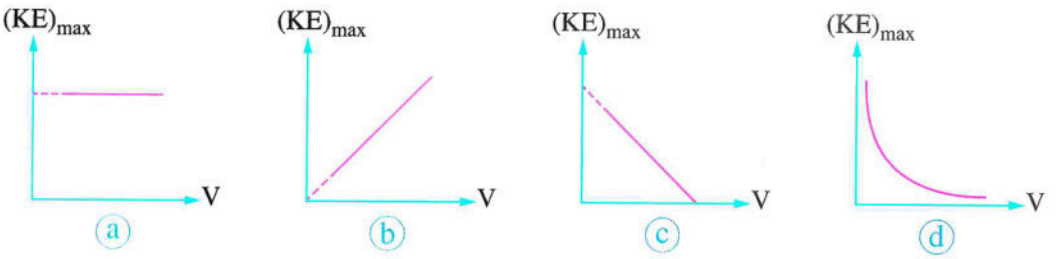


(2)

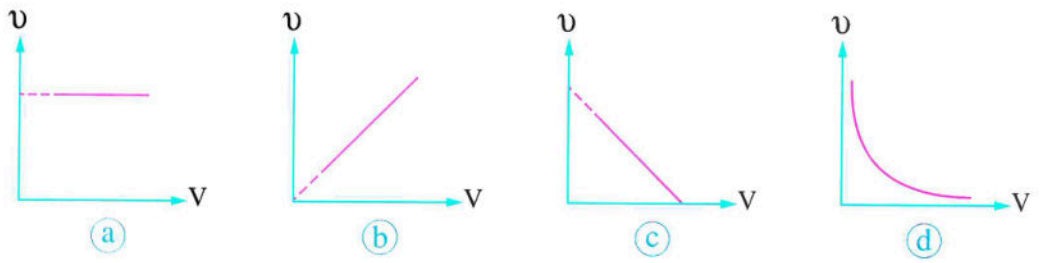
Which of the two photons belongs to the continuous spectrum region of X-rays?

(a) Photon (y).
(b) Photon (z).
(c) Both photons.
(d) Neither photon.

65 Which of the following graphs represents the relation between the maximum kinetic energy $(KE)_{max}$ for the emitted electron from the cathode in Coolidge tube and the potential difference (V) between the cathode and anode?

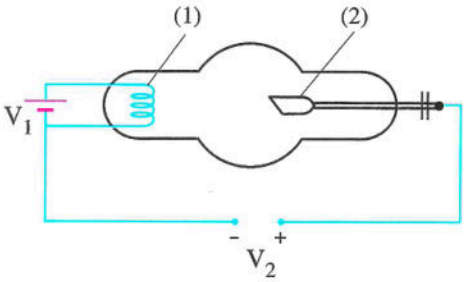


66 Which of the following graphs represents the relation between the maximum frequency (υ) of the emitted photons in the continuous spectrum of X-rays that are generated in Coolidge tube and the potential difference (V) between the cathode and anode?



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67 The opposite figure represents a Coolidge tube, which of the following choices leads to the change of the shortest wavelength of the continuous spectrum of the produced X-rays?



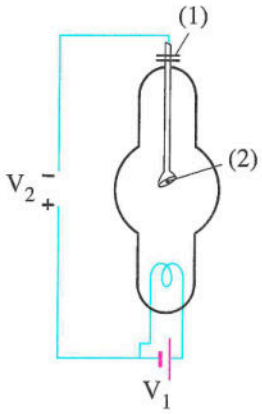
(a) Changing the potential difference V_1.

(b) Changing the potential difference V_2.

(c) Changing the material of component (1).

(d) Changing the material of component (2).

68 The opposite diagram shows a Coolidge tube that failed to produce X-rays although the two voltages V_1 and V_2 were appropriate. For the tube to produce X-rays, ...



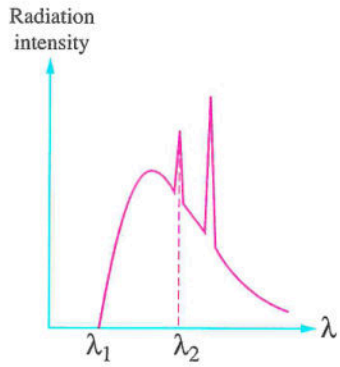
(a) component (1) should be manufactured as a heating coil

(b) component (2) should be manufactured of aluminum

(c) the polarity of the voltage source V_1 should be reversed

(d) the polarity of the voltage source V_2 should be reversed

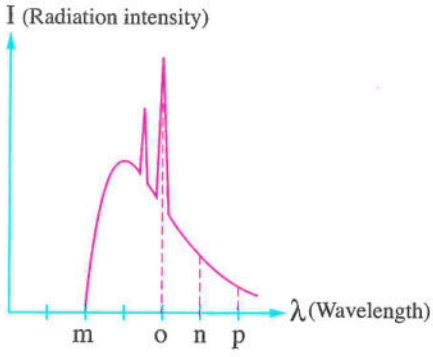
69 The opposite figure represents the relation between the wavelength (λ) for the produced X-rays from Coolidge tube and its intensity. If the potential difference between the filament and target is increased, the value of ..... decreases.



(a) λ_1 only (b) λ_2 only

(c) each of λ_1 and λ_2 (d) neither λ_1 nor λ_2

70 The opposite figure represents the X-ray spectrum which is produced from Coolidge tube. Which wavelength is emitted due to electron transition in the target atom from a higher energy level to an energy level close to the nucleus?

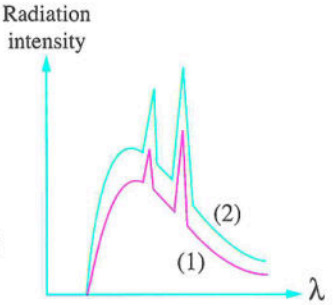


(a) Wavelength (m). (b) Wavelength (o).

(c) Wavelength (n). (d) Wavelength (p).

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71 The opposite graph represents the spectrum of the emitted X-rays from Coolidge tube before and after a particular change as represented in curves (1) and (2) respectively, so this change has occurred due to ..... .



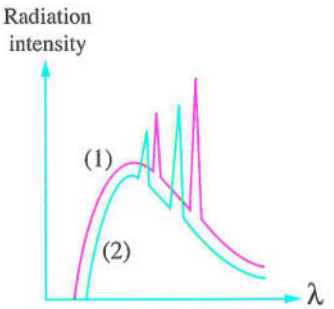
(a) increasing each of the potential difference between the anode and cathode and the atomic number of the target material

(b) decreasing each of the potential difference between the anode and cathode and the atomic number of the target material

(c) increasing the filament current and decreasing the atomic number of the target material

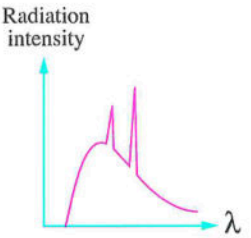
(d) increasing the filament current only

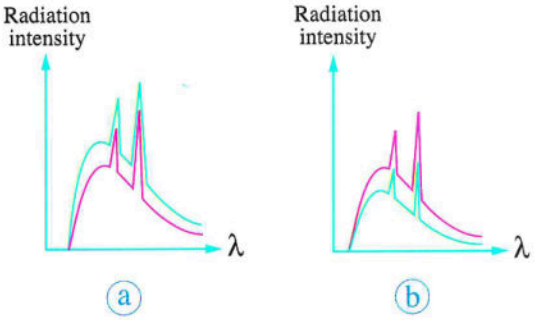
72 The opposite figure shows the relation between the X-rays intensity and its wavelength (λ) for two Coolidge tubes which have targets of two different materials their atomic numbers Z_1 and Z_2 and operate on two different voltages V_1 and V_2 respectively. Thus ..... .



	Relation between V_1, V_2	Relation between Z_1, Z_2
(a)	$V_1 > V_2$	$Z_1 > Z_2$
(b)	$V_1 > V_2$	$Z_1 < Z_2$
(c)	$V_1 < V_2$	$Z_1 = Z_2$
(d)	$V_1 < V_2$	$Z_1 < Z_2$

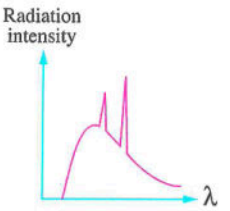
73 The opposite graph represents spectrum of the emitted X-rays from Coolidge tube. Which of the following graphs represents a comparison between that spectrum and the spectrum produced from the tube after increasing the filament current?

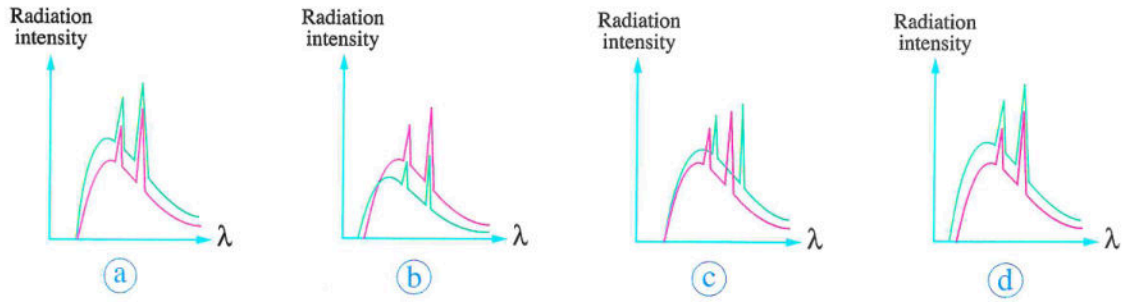




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74 ✱ The opposite graph represents the X-rays spectrum which is produced from a Coolidge tube. Which of the following graphs represents a comparison between that spectrum and the produced one after increasing the potential difference between the cathode and anode in the tube?





75 ✱ If you know that the shortest wavelength of the produced X-rays from Coolidge tube is 0.414 Å, then:

(i) The maximum energy for the photons of X-rays equals ..... .

(a) 1.6×10^{-16} J (b) 4.8×10^{-16} J (c) 1.6×10^{-15} J (d) 4.8×10^{-15} J

(ii) The applied potential difference between the cathode and anode equals ..... .

(a) 10×10^{3} V (b) 28×10^{3} V (c) 30×10^{3} V (d) 48×10^{3} V

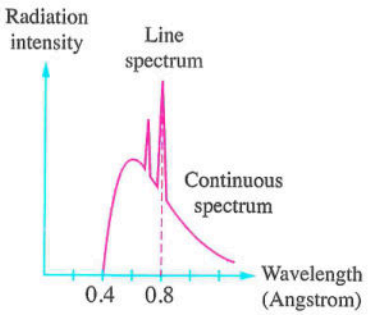
76 ✱ If the potential difference between the anode and the cathode of a tube for generating X-rays is 13255 V, the maximum frequency of the continuous spectrum for these rays is ..... .

(a) 3.2×10^{19} Hz (b) 2×10^{19} Hz (c) 3.2×10^{18} Hz (d) 2×10^{18} Hz

77 ✱ In an X-ray tube, if the maximum kinetic energy of the accelerated electron is 5×10^{-18} J, the shortest wavelength for the generated rays equals ..... .

(a) 1.325×10^{-8} m (b) 3.975×10^{-8} m (c) 1.325×10^{-9} m (d) 3.975×10^{-9} m

78 ✱ The opposite figure represents the emitted X-rays spectrum from Coolidge tube, then:



(i) The potential difference between the filament and the target equals ..... .

(a) 31.05×10^{3} V (b) 3.22×10^{3} V

(c) 9.7×10^{4} V (d) 2.01×10^{4} V

(ii) The maximum frequency of the produced X-rays is ..... .

(a) 3.75×10^{16} Hz (b) 3.75×10^{18} Hz (c) 7.5×10^{16} Hz (d) 7.5×10^{18} Hz

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79 Figure (1) shows the graphical relationship between the intensity of x-rays emitted from a Coolidge tube and the photon energy (E) of these rays.
Figure (2) shows the transitions within one of the atoms of the target material used in the tube, which cause the line spectrum shown in figure (1).

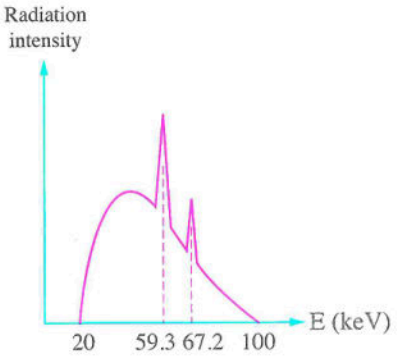


Figure (1)

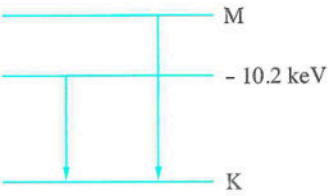


Figure (2)

Using figures (1) and (2), answer the following:

(i) The potential difference between the anode and the cathode in the tube is ..... .

(a) 20 kV (b) 59.3 kV (c) 67.2 kV (d) 100 kV

(ii) The energy levels K, M in the atoms of the target material respectively, are ..... .

(a) −57 keV, −2.3 keV (b) −69.5 keV , −2.3 keV

(c) −49.1 keV, −4.6 keV (d) −69.5 keV , −4.6 keV

80 * If the electric current intensity due to the electrons current in Coolidge tube is 7 mA when the potential difference between the filament and the target is 30 kV, then:

(i) The maximum kinetic energy of the electrons which collide with the target equals ..... .

(a) 4.8×10^{-14} J (b) 4.8×10^{-15} J (c) 4.8×10^{-16} J (d) 4.8×10^{-18} J

(ii) The maximum velocity for the electron at the moment of its arrival to the target is ..... .

(a) 72.36×10^{6} m/s (b) 51.36×10^{6} m/s (c) 10.27×10^{7} m/s (d) 1.1×10^{8} m/s

(iii) The number of electrons which reach the target every second is ..... .

(a) 4.375×10^{13} electrons (b) 4.375×10^{16} electrons

(c) 4.375×10^{19} electrons (d) 4.375×10^{22} electrons

(iv) The shortest wavelength for the produced X-rays is ..... .

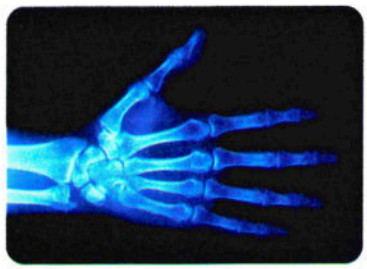
(a) 414 Å (b) 41.4 Å (c) 4.14 Å (d) 0.414 Å

81 In Coolidge tube, if the shortest wavelength of the emitted x-rays is 25 pm, then the wavelength associated with the fastest electrons colliding with the target is approximately ..... .

(a) 4 pm (b) 4.5 pm (c) 5 pm (d) 5.5 pm

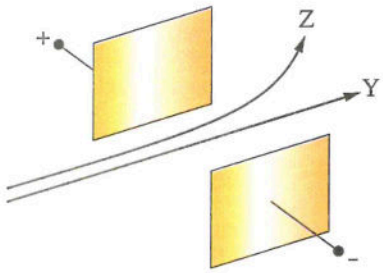
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82 The opposite figure shows an X-ray image of a hand, the usage of X-rays in photographing the bones and detecting fractures' places depends on ..... .



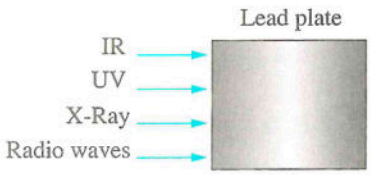
(a) its wide spectral range
(b) its high speed
(c) its short wavelength
(d) its wave nature

83 Two rays Y, Z pass between two parallel charged plates as in the opposite figure, so the two rays respectively are ..... .



(a) cathode ray, X-ray
(b) X-ray, cathode ray
(c) X-ray, laser
(d) laser, cathode ray

84 Four different types of electromagnetic waves are falling on a thick lead plate as in the opposite figure, which of these rays penetrates for longer distance through the lead plate?



(a) Infrared rays. (b) Ultraviolet rays.
(c) X-rays. (d) Radio waves.

Second Essay questions

Bohr's model

1 If the energy of the first level in the hydrogen atom is – 13.6 eV and the radius of the electron orbit in that level is 0.53 Å, according to Bohr's model **calculate:**

(a) The wavelength of the wave associated with the electron in the first level.

(b) The speed of the electron in the first level.

(c) The wavelength of the photon required to excite the electron from the first energy level to the third energy level.

2 **What happens when** hydrogen atoms get excited with different quanta of energies?

3 **Explain why** there is no spectral line in any series of the hydrogen spectrum that has the same wavelength as another spectral line.

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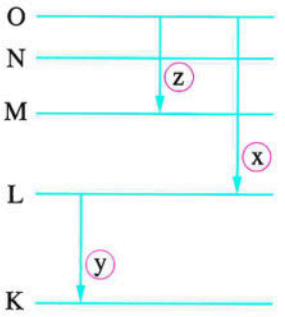
4 A hydrogen atom electron has been excited to a certain level of energy – 0.85 eV and radius 8.48 Å, so according to Bohr's model **calculate:**

(a) The order of the level away from the nucleus.

(b) The wavelength associated with the electron in this level.

(c) The speed of the electron in this level.

5 The opposite diagram represents three possible electron transitions (x), (y) and (z) in a hydrogen atom, **explain why:**



(a) The energy of the photon emitted due to transition (x) is less than energy of the photon emitted due to transition (y).

(b) The two emitted photons due to transitions (y) and (z) are in the invisible light region while the photon emitted due to transition (x) is in the visible light region.

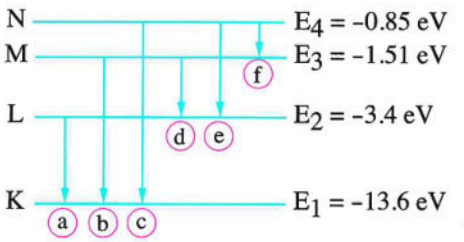
6 **What** could be the smallest equivalent mass to a photon emitted from a hydrogen atom in the Lyman series?

7 **What** is the smallest momentum of a photon that could excite a hydrogen atom from the ground state?

8 In a hydrogen atom, the electron is moving in an energy level of radius r such that its motion is associated with a wavelength $2\pi r$, **calculate** the longest wavelength of a photon that could free this electron from the atom.

9 If the longest wavelength in Lyman series is λ, **find** the longest wavelength in the Pfund series in terms of λ.

10 The opposite diagram represents six electron transitions that occurred in a hydrogen atom from higher energy levels to lower energy levels. Assuming the emitted photons fell on the surface of the cathode of a photoelectric cell with a work function of 2.14 eV:



(a) **How many** electrons could be released from the cathode?

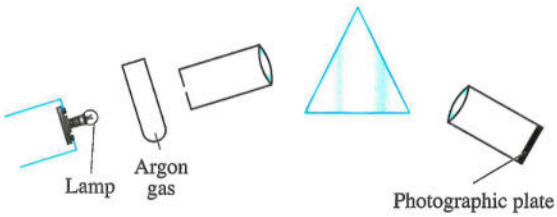
(b) **Which** photons causes an electron emission with the highest speed?

The spectrometer and the spectra

11 **Compare between** the linear absorption spectrum and the linear emission spectrum for an element.

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12 The opposite diagram illustrates the use of a spectroscope in spectral analysis, so when operating the light source, **describe the resulted spectrum in the case of** :



(a) The presence of an argon gas tube between the lamp and the spectroscope.

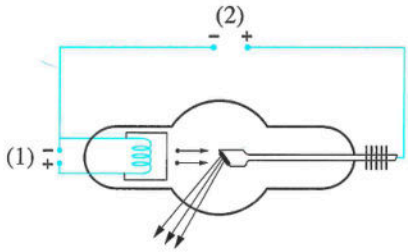
(b) The absence of the argon gas tube from between the lamp and the spectroscope.

13 **How can** a spectroscope be used to distinguish between two different elements?

14 **Explain how** the study of atomic spectra helped in the discovery of the composition of the solar atmosphere.

X-rays

15 The opposite figure illustrates the structure of Coolidge tube:



(a) **Compare between** the magnitudes of voltages (1) and (2) and the role of each of them.

(b) **Clarify how** energy transformations that occur in the tube take place to produce X-rays?

(c) **What** is the phenomenon responsible for electrons emission from the cathode? **And why** do these electrons get emitted?

(d) **What are the characteristics of the used material in:**

① The target. ② The filament.

16 In a Coolidge tube:

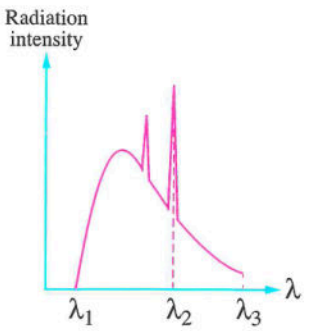
(a) **Why** is the anode made of copper and equipped with a cooling fin?

(b) **How** can you change the ability of the produced X-rays to penetrate through matter?

(c) **How** can you change the intensity of the produced X-rays?

17 **Explain why** the characteristic line spectrum of X-rays is not emitted in a Coolidge tube unless the potential difference between the filament and the target exceeds a certain value.

18 The opposite graph represents the X-rays spectrum emitted from a Coolidge tube, **explain:**



(a) The occurrence of the continuous range of wavelengths from λ_1 to λ_3

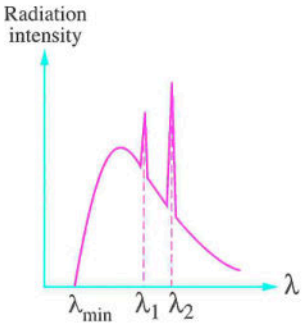
(b) The probable reason for the occurrence of the X-rays spectrum whose wavelength is:

① λ_1 ② λ_2

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19 The opposite graph represents the X-rays spectrum, given that the characteristic spectrum of X-rays that are produced due to the transition of the electrons in the target material from levels n = 2, n = 3 to level n = 1, **then determine with explanation which of the two wavelengths λ_1 and λ_2 gets produced from the transition from:**

(a) n = 2 to n = 1 (b) n = 3 to n = 1

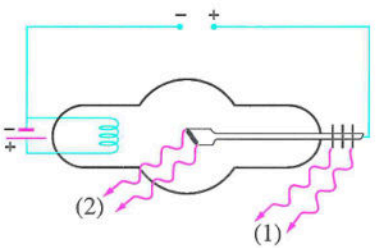


20 The opposite diagram shows a Coolidge tube producing two different types of electromagnetic radiation (1) and (2).

Which of them:

(a) Has photons with greater momentum?

(b) Is used to determine the crystal structure of solids?



21 **What are the types of** the input energy and the output energies from a Coolidge tube? **And which** part of these energies get wasted?

22 **Give reasons:** Coolidge tube produces electromagnetic radiation of very high frequencies.

23 In a Coolidge tube, if the maximum momentum of electrons upon impact with the target is 63.7×10^{-25} kg.m/s, **calculate** the shortest wavelength of the emitted X-rays.

24 An X-ray tube operates at a potential difference of 40 kV, if the electron current through the tube is 5 mA, **calculate:**

(a) The number of electrons colliding with the target per second.

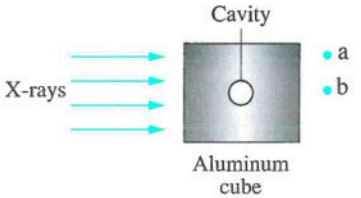
(b) The rate of electrical energy consumed in the tube.

(c) The rate of X-ray energy produced if the tube efficiency is 2%.

25 **What are the results of** passing X-rays through a sample of gas?

26 **Explain why** X-rays are used instead of visible light in airports and seaports for checking bags contents.

27 X-rays are set to get incident on an aluminum cube containing a gas cavity as shown in the figure, **compare between** the intensities of X-rays at points a and b, **then explain** how this can be used to detect internal structure defects in metals.

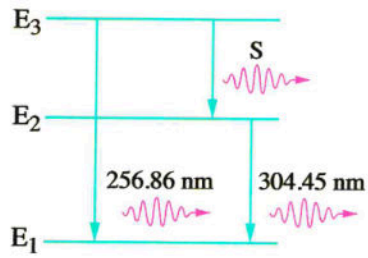


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Questions of the Previous Years Exams on Chapter 6

Bohr's model

1 The opposite figure illustrates an excited atom that produces wavelengths due to the transition of an electron from a higher energy level to a lower energy level, then the wavelength (S) equals ..... . *(Exp. 23)*



(a) 2250 nm (b) 1643 nm

(c) 3000 nm (d) 450 nm

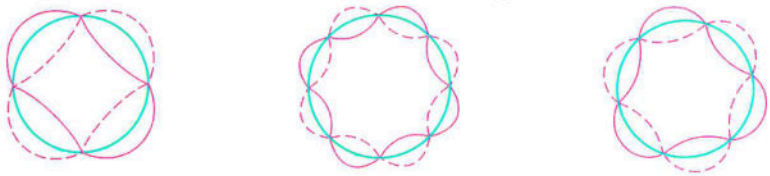
2 A photon hits an electron in the ground state of hydrogen atom, as a result the electron moved to excitation level (N), which of the choices expresses the incident photon's wavelength? *(1st session 22)*

(a) 1.56×10^{-26} m (b) 1.56×10^{-8} m (c) 9.74×10^{-26} m (d) 9.74×10^{-8} m

3 The longest wavelength of the visible spectrum emitted from the hydrogen atom approximately equals ..... . *(2nd session 23)*

(a) 6760 Å (b) 5670 Å (c) 6576 Å (d) 7570 Å

4 The following figures show the standing waves associated with the motion of the electron in three energy levels X , Y , Z for H-atom according to Bohr's atomic model.



(X) (Y) (Z)

Which choice is correct? *(1st session 24)*

(a) A photon in the visible region is produced when the electron transferred from level (Y to Z).

(b) The energy of level Z is less than that of level X.

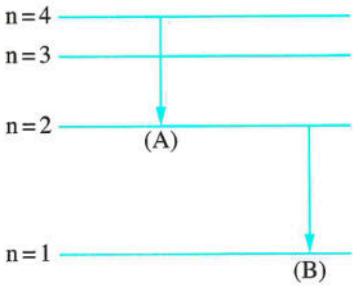
(c) The difference in energy between levels (Z and X) is greater than that of (Z and Y).

(d) The energy of level X is greater than that of level Y.

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5 The figure shows some transitions of the electron between energy levels for H-atom.

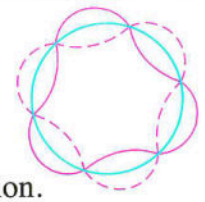
So, the ratio between $\left(\frac{\upsilon_A}{\upsilon_B}\right)$ = ... *(1st session 24)*



(a) $\frac{4}{1}$ (b) $\frac{1}{4}$

(c) $\frac{2}{1}$ (d) $\frac{1}{2}$

6 According to Bohr's model of the hydrogen atom and from the shown figure, which choice is correct when the electron returns from higher energy levels to that level in the figure? *(2nd session 24)*

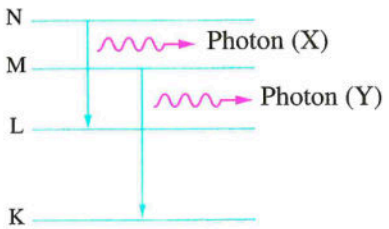


(a) Emitting photon in UV region. (b) Emitting photon in IR region.

(c) Emitting photon in visible region. (d) Emitting photon in X-rays region.

7 The opposite figure shows two possible transitions of an electron in an excited hydrogen atom , so the ratio between: *(2nd session 24)*

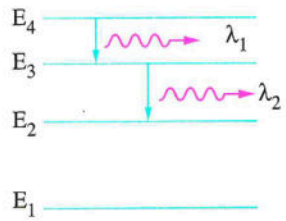
$\frac{\text{The momentum of photon (X)}}{\text{The momentum of photon (Y)}}$ = ...



(a) $\frac{55.5}{148}$ (b) $\frac{128}{7}$

(c) $\frac{27}{128}$ (d) $\frac{148}{55.5}$

8 The opposite figure shows two possible transitions of electron in hydrogen atom, so the ratio between the wavelengths of the two corresponding photons $\left(\frac{\lambda_1}{\lambda_2}\right)$ equals ..... . *(Practice 25)*



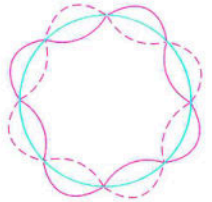
(a) $\frac{20}{7}$ (b) $\frac{7}{20}$

(c) $\frac{9}{4}$ (d) $\frac{27}{5}$

9 Based on your study, the longest wavelength in the five spectral series of the hydrogen atom, occurs when the excited electron returns from level ..... . *(Practice 25)*

(a) ∞ to level one (b) ∞ to level five (c) six to level five (d) two to level one

10 The opposite figure represents the standing wave associated with the motion of an electron in a hydrogen atom at one of the energy levels of atom, then the energy at this level equals ..... *(Practice 25)*



(a) –13.6 eV (b) –3.4 eV

(c) – 0.544 eV (d) – 0.85 eV

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11 A group of hydrogen atoms has been excited such that the electron in each atom is in the fifth energy level and the electron can make a transition between any two of these energy levels, then the number of possible spectral series that can be emitted from the atom is ..... .

(Practice 25)

(a) 4 (b) 6 (c) 8 (d) 10

12 Figure (1) shows the energy level diagram of an excited atom that emits specific wavelengths as the electron make transitions from a higher energy level to a lower one, and figure (2) shows these wavelengths.

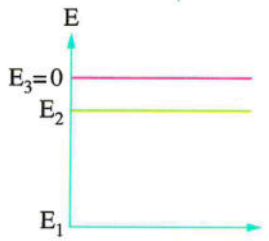


Figure (1)

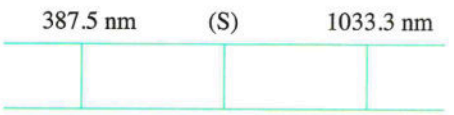


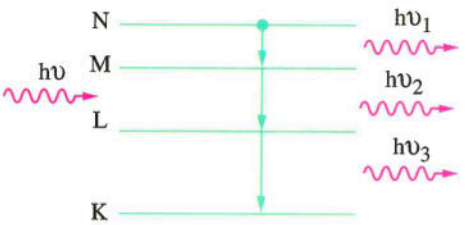
Figure (2)

Then wavelength (S) equals ..... .

(Practice 25)

(a) 620 nm (b) 775 nm (c) 516.7 nm (d) 710.4 nm

13 An electron in hydrogen atom was at energy level K, it absorbed a photon with energy (E) and frequency (υ), causing it to make a transition to level N, and after a certain amount of time, the electron returned back to level K passing through levels M and L, as shown in the diagram such that during electron transition:



- From level N to level M, a photon of frequency (υ_1) was emitted.
- From level M to level L, a photon of frequency (υ_2) was emitted.
- From level L to level K, a photon of frequency (υ_3) was emitted.

Which of the following equations correctly describes the relation between the frequency of the absorbed photon and the frequencies of the emitted photons ?

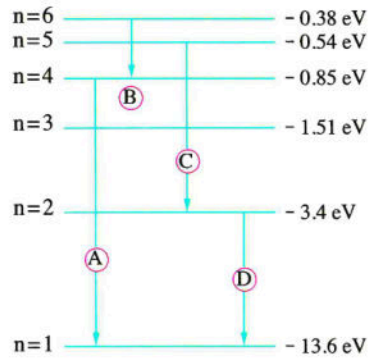
(Practice 25)

(a) $\upsilon > (\upsilon_1 + \upsilon_2 + \upsilon_3)$ (b) $\upsilon = \upsilon_1 + \upsilon_2 + \upsilon_3$

(c) $\upsilon < (\upsilon_1 + \upsilon_2 + \upsilon_3)$ (d) $\upsilon = \upsilon_2 + \upsilon_3 - \upsilon_1$

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14 The diagram shows some possible transitions for the electron between energy levels in the hydrogen atom. Which of the transitions shown in the figure results in the emission of a photon with a wavelength of 974 Å?



(a) Transition C

(b) Transition B

(c) Transition D

(d) Transition A

(1st session 25)

15 ✳ In an atom, as an electron transfers from the third energy level (M) to the second energy level (L), a photon of energy 3×10^{-19} J is emitted. And as it transfers from the second energy level (L) to the first energy level (K), a photon of energy 1.634×10^{-18} J is emitted. What is the frequency of the emitted spectrum as the electron transfers from the third energy level (M) to the first energy level (K)? *(2nd session 25)*

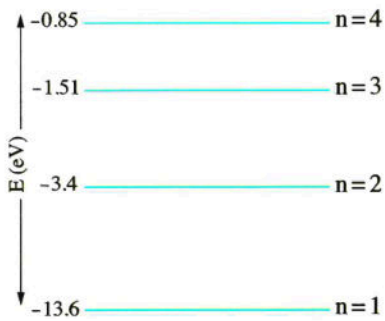
(a) 2.92×10^{15} Hz

(b) 4.54×10^{14} Hz

(c) 2.47×10^{14} Hz

(d) 2.01×10^{15} Hz

16 The figure shows a diagram of the energy levels of an excited hydrogen atom in the fourth energy level. From the figure, ... *(Practice 26)*



(a) the lowest frequency of photons that the atom can emit in this state in the visible-light region.

(b) the highest frequency of photons that the atom can emit in this state is 1.6×10^{14} Hz.

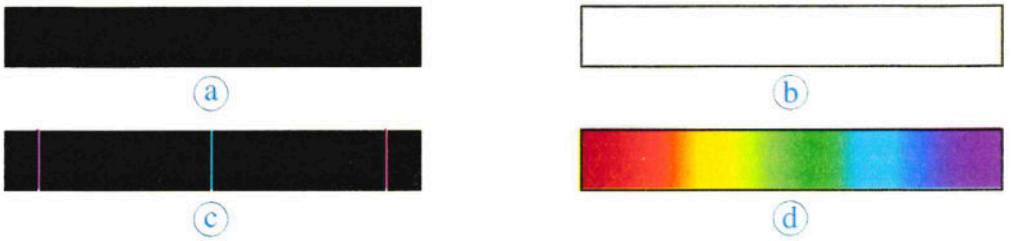
(c) the shortest wavelength of visible light photons that the atom can emit in this state is 487.1 nm.

(d) the shortest wavelength of photons that the atom can emit in this state in the infrared region.

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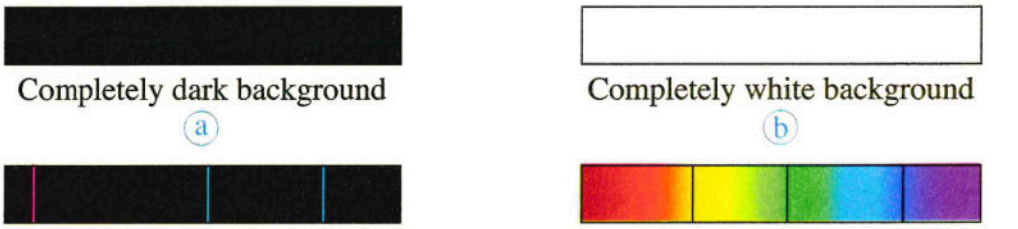
The spectrometer and the spectra

17 Which figure represents the spectrum resulting from an excited hydrogen gas? *(Exp. June 21)*



ⓐ ⓑ ⓒ ⓓ

18 Which figure shows the spectrum produced, when a white light spectrum passes through a gas? *(2nd session 21)*



Completely dark background ⓐ

Completely white background ⓑ

Dark background with bright lines ⓒ

Colored background with dark lines ⓓ

19 While looking through the eye piece of each spectrometer, we see ..... . *(1st session 22)*

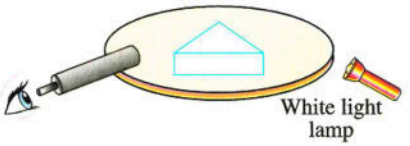


Figure (1)

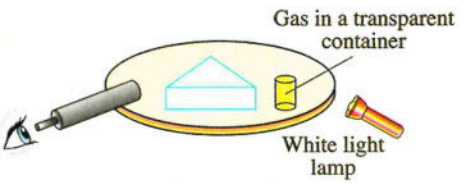
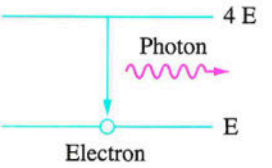


Figure (2)

	In figure (1)	In figure (2)
ⓐ	line absorption spectrum	line emission spectrum
ⓑ	line emission spectrum	continuous spectrum
ⓒ	continuous spectrum	line absorption spectrum
ⓓ	line absorption spectrum	continuous spectrum

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20 The opposite figure illustrates an electron transition between two energy levels of an atom producing a photon, then ..... . *(2nd session 22)*



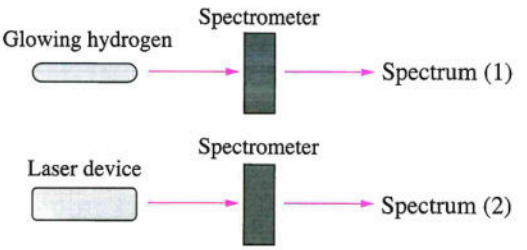
	Spectrum type	Photon energy
(a)	line absorption	3 E
(b)	line emission	3 E
(c)	continuous	5 E
(d)	line emission	5 E

21 From the figure, spectra (1) and (2) respectively are ....., ... *(1st session 23)*

(a) continuous, continuous

(b) continuous, linear

(c) linear, linear (d) linear, continuous

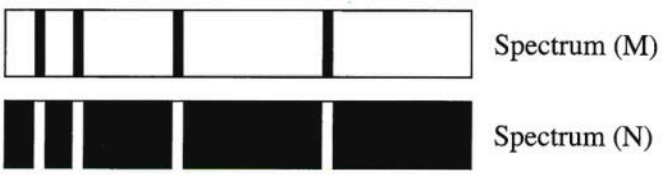


22 Fraunhofer lines in the solar spectrum represent ..... . *(Practice 25)*

(a) line emission spectrum (b) line absorption spectrum

(c) continuous emission spectrum (d) monochromatic spectrum

23 ✳ The figure shows two spectra (M) and (N) received separately on the photographic plate of a spectrometer. *(1st session 25)*



(Note: The colored portions appeared on the photographic plate are represented in white in this figure)

The two spectra are .....

(a) for two different gases, both are absorption spectra

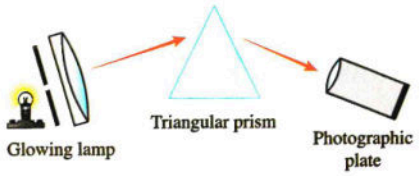
(b) for the same gas, where spectrum (M) is emission spectrum and spectrum (N) is absorption spectrum

(c) for two different gases, both are emission spectra

(d) for the same gas, where spectrum (N) is emission spectrum and spectrum (M) is absorption spectrum

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24 The diagram shows a parallel beam of light emitted by a glowing tungsten lamp. The beam passes through a triangular prism and then falls onto a sensitive photographic plate.



What happens to the spectrum formed on the photographic plate when a sample of a cold gas is placed between the lamp and the prism on the path of the lamp light?

(a) The continuous spectrum of the lamp disappears, and black lines characteristic to the gas type appear.

(b) The continuous spectrum of the lamp disappears, and colored spectral lines characteristic to the gas type appear.

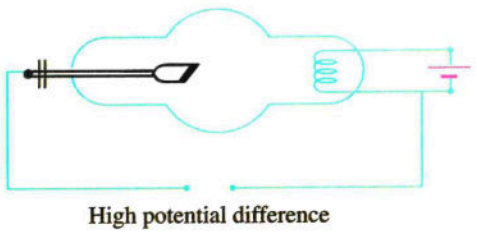
(c) Colored spectral lines characteristic to the gas type appear in addition to the continuous spectrum of the lamp.

(d) Black lines characteristic to the gas type appear throughout the continuous spectrum of the lamp.

(2nd session 25)

X-rays

25 In Coolidge tube in the figure, to produce X-rays the atomic number of the target material was 42, so to produce a longer wavelength for the characteristic X-rays radiation, we must change the target material to another element of atomic number ..... .



(Exp. June 21)

(a) 29 (b) 74 (c) 82 (d) 55

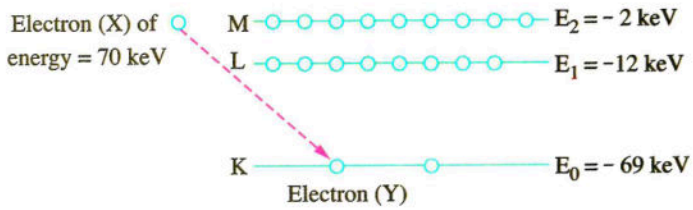
26 In Coolidge tube, if the velocity of the electrons when colliding with the target is 7.34×10^6 m/s, so the minimum wavelength of the produced X-ray spectrum equals ..... .

(Exp. June 21)

(a) 8.11 nm (b) 0.811×10^{-9} m (c) 0.059 nm (d) 5.9×10^{-10} m

27 The following figure shows some of the energy levels of the tungsten atom that is used as a target in Coolidge tube. If the collision between electron (X) and electron (Y) leads to the ejection of electron (Y) out of its atom, which one of the following probabilities is considered the energy of photons of the resulting characteristic X-ray spectrum?

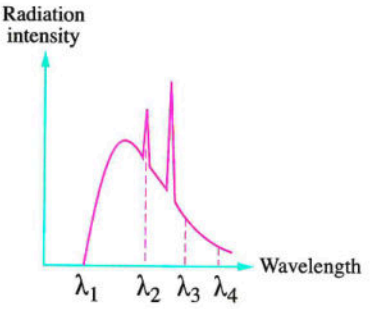
(1st session 21)



(a) 69 keV, 70 keV (b) 14 keV, 68 keV (c) 1 keV, 72 keV (d) 67 keV, 57 keV

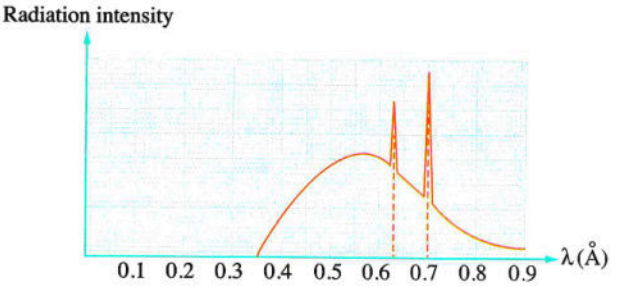
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28 The opposite graph represents the relation between intensity of radiation and wavelength for X-ray spectrum, so the wavelength which decreases as the atomic number of the target material increases, is ..... . *(1st session 21)*



(a) λ_2 (b) λ_4
(c) λ_1 (d) λ_3

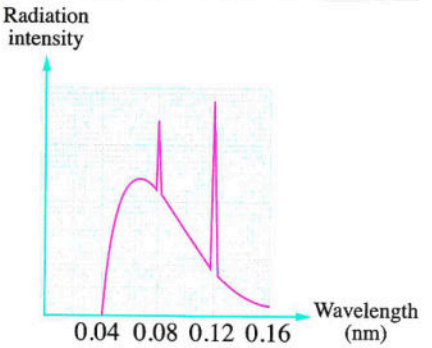
29 The graph represents the relation between the radiation intensity and wavelength of the produced X-ray spectrum from Coolidge tube. From the graph:



The ratio : $\frac{\text{The least frequency of the characteristic spectrum}}{\text{The greatest frequency of the continuous spectrum}}$ = ... *(2nd session 21)*

(a) 0.58 (b) 1.75 (c) 2 (d) 0.5

30 The figure represents the relation between intensity of radiation and wavelength for the X-ray spectrum, so the wavelength of the characteristic X-ray which corresponds to the maximum momentum for its photons is ..... . *(2nd session 21)*

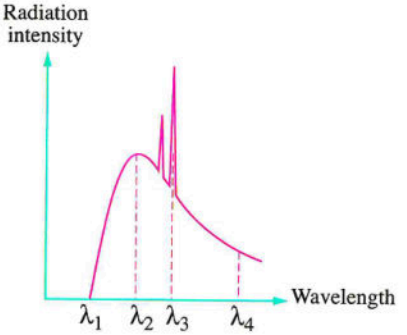


(a) 0.04 nm (b) 0.08 nm
(c) 0.12 nm (d) 0.16 nm

31 When using a certain element as a target material in Coolidge tube to produce X-rays, photon of frequency 5.43×10^{18} Hz is produced when an excited atom is transferred between two energy levels where the energy of one of them is (– 1.5 keV). So, the energy of the other one equals ..... . *(1st session 22)*

(a) – 24 keV (b) – 22.5 keV (c) – 27 keV (d) – 25.5 keV

32 The opposite graph represents the relation between radiation intensity and wavelength for the X-ray spectrum. So, the wavelength resulting from the transition of electrons inside an excited atom of the target from a higher energy level (E_2) to another lower energy level (E_1) is ..... . *(1st session 22)*



(a) λ_1 (b) λ_3
(c) λ_2 (d) λ_4

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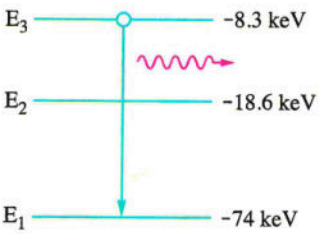
33 In Coolidge tube to generate X-rays, if an electron is ejected towards the target with energy 70 keV that gets decreased to 54.5 keV due to scattering, the wavelength of the continuous spectrum photon of the X-rays produced in this case equals ..... .

(2nd session 22)

(a) 1.01×10^{-11} m (b) 2.28×10^{-11} m (c) 8.01×10^{-11} m (d) 8.77×10^{-11} m

34 The figure illustrates the values of some energy levels in an element used as a target in Coolidge tube. When electron transferred as in the figure, the wavelength of the produced X-ray photon = ...

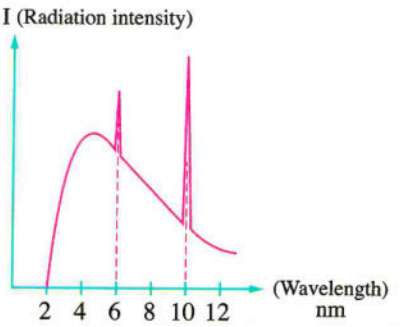
(2nd session 22)



(a) 9×10^{-10} m (b) 3.6×10^{-11} m
(c) 6×10^{-10} m (d) 1.9×10^{-11} m

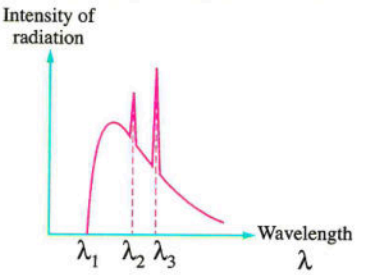
35 The smallest characteristic wavelength of X-rays from the opposite figure equals .....

(Exp.23)



(a) 8 nm
(b) 10 nm
(c) 4 nm
(d) 6 nm

36 The figure represents the relation between the intensity of X-ray radiation produced from a Coolidge tube operated at potential difference (V) and its wavelength. On increasing both of the filament current intensity and the potential difference between the cathode and the anode. Which choice is the correct answer? *(1st session 23)*



	λ_1	λ_2	λ_3	**Radiation intensity**
(a)	Increases	Does not change	Does not change	Decreases
(b)	Decreases	Increases	Does not change	Does not change
(c)	Decreases	Does not change	Does not change	Increases
(d)	Increases	Does not change	Does not change	Increases

37 The scientific principle of using X-rays to study the crystalline structure of solids depends on ..... .

(2nd session 23)

(a) their ability to diffract when passing through crystals
(b) the intensity of X-rays
(c) the wide spectrum of X-rays
(d) the quantum nature of X-rays

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38 In Coolidge tube, tungsten $_{(74)}W$ was used as a target to produce X-rays, so that one of the characteristic wavelengths of X-ray radiation was 1.8×10^{-11}m, so on replacing the target with another one made of molybdenum $_{(42)}Mo$, one of the characteristic wavelengths of X-rays radiation probably equals ..... . *(2nd session 23)*

(a) 1.5×10^{-3} nm (b) 4.9×10^{-3} nm (c) 2.8×10^{-4} nm (d) 7.1×10^{-2} nm

39 Figure (1) graphically represents the spectrum of X-rays emitted from a Coolidge tube, while figure (2) shows some electron transitions between energy levels in the target atom.

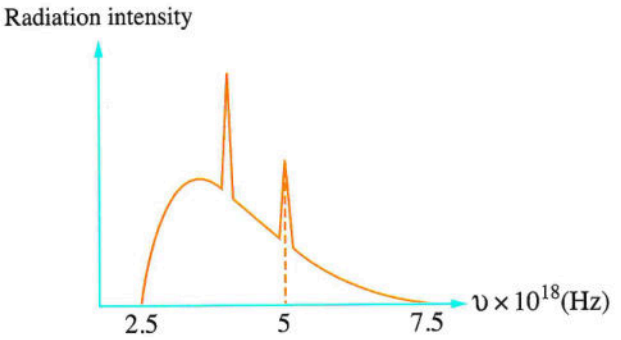


Figure (1)

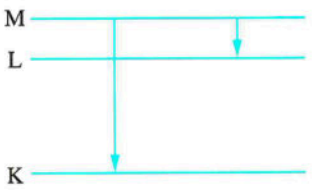


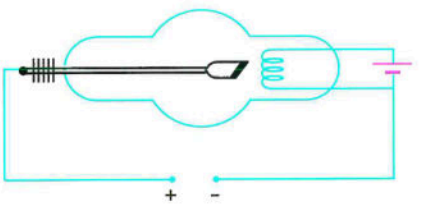
Figure (2)

Which choice is correct for the frequency of the characteristic photons and their transitions? *(1st session 24)*

(a) 5×10^{18} Hz, from level (M) to level (K). (b) 5×10^{18} Hz, from level (M) to level (L).

(c) 5.3×10^{18} Hz, from level (M) to level (K). (d) 5.3×10^{18} Hz, from level (M) to level (L).

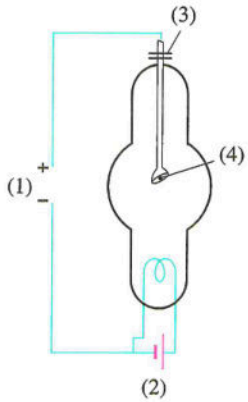
40 In Coolidge tube as shown, if the target material was made of an element, its atomic number = 42, then the experiment is repeated by another target material made of an element, its atomic number = 76, and by increasing the potential difference between the terminals of Coolidge tube so, which of the following choices is correct? *(2nd session 24)*



	The wavelength of characteristic radiation	The smallest wavelength of the continuous radiation
(a)	Increases	Increases
(b)	Decreases	Decreases
(c)	Decreases	Increases
(d)	Increases	Decreases

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41 The opposite figure represents a diagram of Coolidge tube for generating x-rays, the component that is responsible for accelerating the electrons emitted from the filament is ..... . *(Practice 25)*



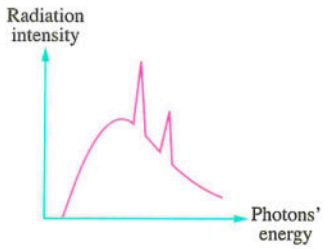
(a) (1)

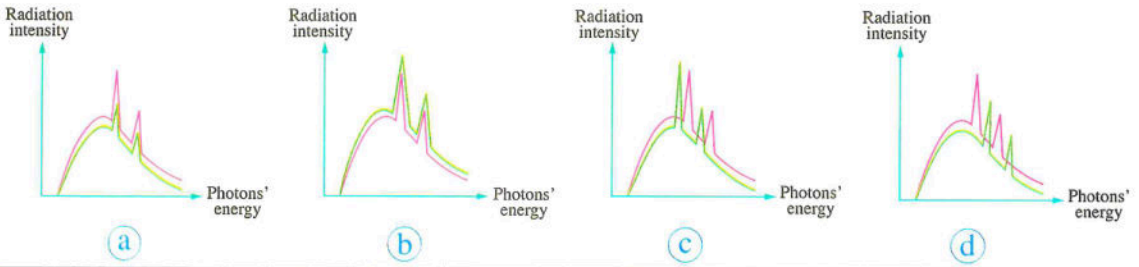
(b) (2)

(c) (3)

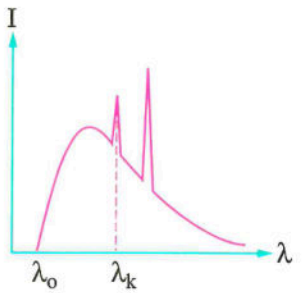
(d) (4)

42 The opposite graph shows the relation between the X-ray intensity and the energy of the photons produced by a Coolidge tube, if the target material is replaced with another of lower atomic number, while keeping the voltage between the cathode and anode constant, which of the following graphs can represent the resulting spectrum ? *(Practice 25)*





43 The graph shows the relationship between the intensity of the X-rays (I) and the wavelength (λ) emitted by Coolidge tube. When the potential difference between the target and the filament is increased, the difference between the values of (λ_o) and (λ_k) .....



(a) becomes equal to the value of λ_o

(b) decreases

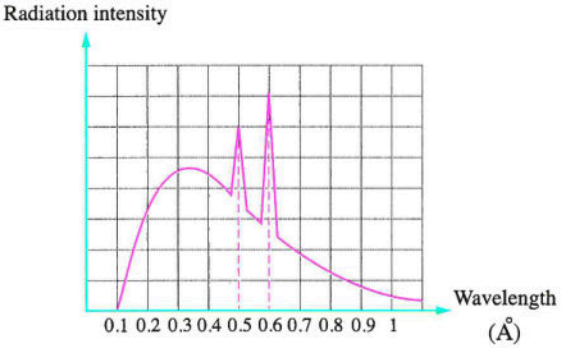
(c) remains unchanged

(d) increases

(1st session 25)

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44 The graph represents the relationship between the intensity (I) and wavelength (λ) of X-rays emitted in Coolidge tube. Which wavelengths in the figure represent the characteristic spectrum of the target material?

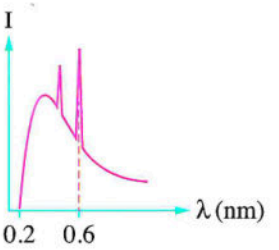


(a) 0.1 Å, 0.5 Å (b) 0.6 Å, 0.5 Å

(c) 0.1 Å, 0.6 Å (d) 0.6 Å, 1 Å

(2nd session 25)

45 The opposite graph represents the relation between the intensity of radiation (I) and the wavelength (λ) of X-rays emitted from Coolidge tube, **calculate:**



(1) The maximum energy of the emitted photons.

(2) The energy of one of the emitted photons for linear X-ray spectrum.

.....

(Exp. 23)

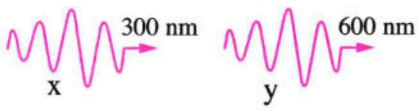
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Test on Chapters 5 and 6

Answered

First **Choose the correct answer:**

1 The opposite figure represents two photons, x and y, of two spectra of different wavelengths propagate in air, then the ratio between the speeds of the two photons $\left(\frac{v_x}{v_y}\right)$ is ..... .



(a) $\frac{4}{1}$ (b) $\frac{2}{3}$ (c) $\frac{1}{1}$ (d) $\frac{1}{3}$

2 Figure (1) represents the emitted spectrum from the gas of one element, so what does figure (2) represent?



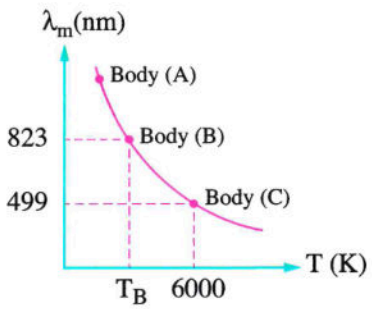
Figure (1)



Figure (2)

(a) The continuous spectrum for the same element.
(b) The emission spectrum for another element.
(c) The absorption spectrum for the same element.
(d) The absorption spectrum for another element.

3 The opposite graph represents the relation between the wavelength (λ_m) associated with the maximum radiation intensely emitted from a number of glowing objects and the absolute temperature (T) for each of them, so the temperature of body (B) equals ..... .

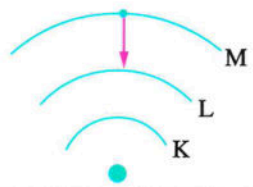


(a) 5895.8K (b) 4622.8K
(c) 3637.9K (d) 3364.9K

4 When an X-rays photon of a wavelength 1.2×10^{-12} m collides with an electron, the photon is scattered with a frequency of 1.5×10^{20} Hz, so the acquired kinetic energy by the electron will be ..... . ($h = 6.625 \times 10^{-34}$ J.s, $c = 3 \times 10^{8}$ m/s)

(a) 2.955×10^{-19} J (b) 1.257×10^{-17} J (c) 8.752×10^{-16} J (d) 6.625×10^{-14} J

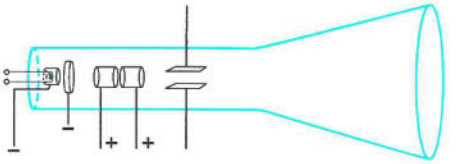
5 When the electron of the hydrogen atom moves as shown in the figure, the atom radiates a photon of ..... .



(a) red rays (b) violet rays
(c) IR rays (d) UV rays

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6 The opposite diagram illustrates the structure of a cathode ray tube with a missing part of it, then the figure that shows what will be displayed on the screen as a result of the Collision of electron beam with it is ..... .



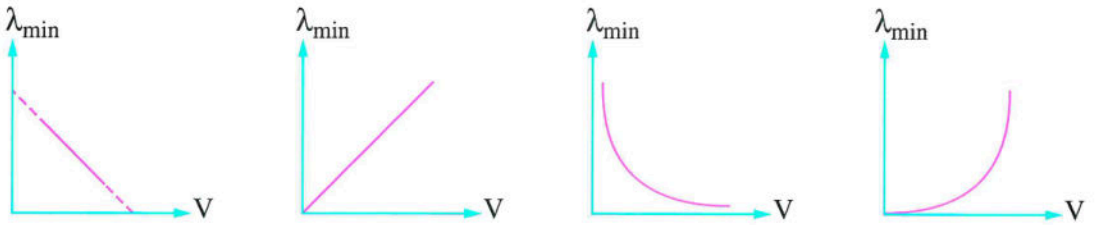


(a) (b) (c) (d)

7 If the radius of energy level L in the hydrogen atom is r, the momentum of the electron in this level according to Bohr's model equals ..... .

(a) πhr (b) $\frac{\pi r}{h}$ (c) $\frac{h}{\pi r}$ (d) $\frac{h}{2\pi r}$

8 Which of the following graphs represents the relation between the shortest wavelength (λ_{min}) of X-ray produced from Coolidge tube and the potential difference (V) between the filament and the target?

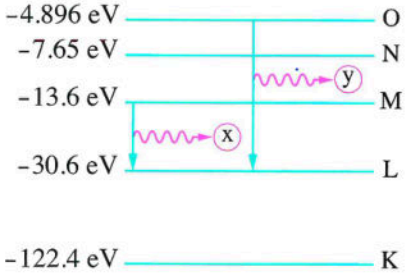


(a) (b) (c) (d)

9 A star emits energy at a rate 4.5×10^{25} J/s in space, so the average mass that the star loses in one minute equals ..... . (Given that: $c = 3 \times 10^8$ m/s)

(a) 3×10^8 kg (b) 5×10^8 kg (c) 3×10^{10} kg (d) 5×10^{10} kg

10 Two photons (x) and (y) are emitted due to the transition of electrons between energy levels in the atom of an element as in the opposite figure, so the ratio between the equivalent mass of the two photons $\left(\frac{m_x}{m_y}\right)$ equals ..... .

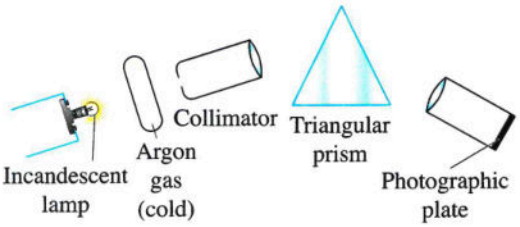


(a) 0.22 (b) 0.33

(c) 0.66 (d) 0.99

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11 A spectroscope is used in spectral analysis produced from an incandescent lamp twice, once when its spectrum falls directly on the spectroscope and another time in the presence of a tube containing cold argon gas between the lamp and the spectroscope as shown in the opposite figure, so the two resulted spectra on the photographic plate represent, respectively, ... , ...



(a) continuous spectrum, continuous spectrum
(b) continuous spectrum, line absorption spectrum
(c) line absorption spectrum, line absorption spectrum
(d) line absorption spectrum, continuous spectrum

12 In the solar radiation, if the photon emission rate for one of the wavelengths of ultraviolet ray is X_1 and the photon emission rate for one of the wavelengths of blue light is X_2, the ratio $\left(\frac{X_1}{X_2}\right)$ is ..... .

(a) greater than 1 (b) less than 1 (c) equal to 1 (d) indeterminable

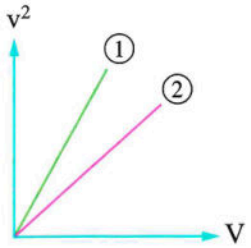
13 An infrared photon while traveling in space, compared to a red radiation photon, has ..... .

(a) a smaller momentum
(b) a shorter wavelength
(c) a higher energy
(d) a higher speed

14 The use of X-rays in studying the crystalline structure of solids is because of having ..... .

(a) large wavelength
(b) high speed
(c) the ability to be ionized
(d) the ability to be diffracted between the atoms

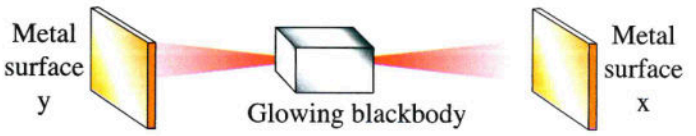
15 Suppose there are two particles ① and ② with the same electric charge (q) and each of them is accelerated several times, each time under a different potential difference, the opposite graph represents the relation between the square of the speed (v^2) of the two particles and the potential difference (V) affecting them, **which** of the two particles has a greater mass?



(a) Particle ①.
(b) Particle ②.
(c) The two particles have the same mass.
(d) The answer in indeterminable.

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16 A glowing blackbody placed in the middle of the distance between two metal surfaces (x and y) having the same dimension as in the opposite figure, if the work function of the metal (x) was double the work function of the metal (y) and electrons were emitted from the two surfaces, **from which** surface are electrons emitted at a greater rate?



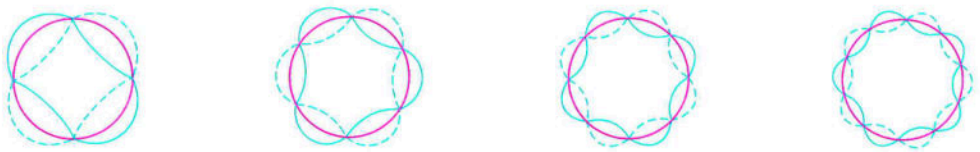
(a) Surface (x).
(b) Surface (y).
(c) They are emitted from the two surfaces with the same rate.
(d) The answer is indeterminable.

17 Two (light) beams have the same power, each of them has a single wavelength, one is of yellow light and the other of blue light, both of them get incident on surfaces of the same metal so that electrons are emitted in both cases by rate $(\phi_L)_1$,$(\phi_L)_2$ respectively, assuming that every photon makes an electron get emitted, **which of the following relations is correct?**

(a) $(\phi_L)_1 > (\phi_L)_2$
(b) $(\phi_L)_1 < (\phi_L)_2$
(c) $(\phi_L)_2 = (\phi_L)_1$
(d) The answer is indeterminable.

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In hydrogen atom, the electron is excited to an orbit of radius 8.48 Å and the speed of the electron in the orbit is 5.47×10^5m/s, which of the following diagrams represents the standing wave which is associated with the motion of the electron in this orbit?
(Given that: $h = 6.625 \times 10^{-34}$ J.s, $m_e = 9.1 \times 10^{-31}$ kg)



(a) (b) (c) (d)

19 If electrons are accelerated by the electron microscope under a potential difference of 25 kV, the wavelength that is associated to electrons is λ, then accelerated another time under potential difference of 6.25 kV, the wavelength of the wave that is associated with the motion of electrons becomes ..... .

(a) 4λ (b) 2λ (c) $\frac{\lambda}{2}$ (d) $\frac{\lambda}{4}$

Second Answer the following question

20 Two photons have fallen on the surface of a metal each at a time, such that the momentum of the first is 1.6×10^{-27} kg.m/s and that of the second is 2.4×10^{-27} kg.m/s. If the work function of the metal is 2.5 eV, **calculate** ratio between the maximum possible speeds of the emitted electrons from the surface in both cases $\left(\frac{v_1}{v_2}\right)$.

(Given that: $c = 3 \times 10^8$ m/s, $e = 1.6 \times 10^{-19}$ C)

.....

.....

.....

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